Lecture 8

Root strings and rank-one subalgebras

Objectives. *Each root α determines a copy of $\mathfrak{sl}_2$ inside $\mathfrak{g}$. We apply Lecture 6 to $\mathfrak{g}$ as a module for this copy.*

By the end of this lecture you will be able to:

- Construct the root triple of a root and compute its coroot in a matrix algebra.
- Prove that root spaces are lines, that $\pm\alpha$ are the only roots proportional to α, and that $\beta(h_\alpha) \in \mathbb{Z}$ for all roots α, β.
- Determine a root string and compute a Cartan entry from it.
- Explain why the real span of the roots is Euclidean, while the Killing form of $\mathfrak{g}$ is not.

8.1 Input from Lecture 7

Throughout, $\mathfrak{g}$ is a semisimple Lie algebra over $\mathbb{C}$ and κ is its Killing form. Fix a Cartan subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, that is, a maximal toral subalgebra. For $\lambda \in \mathfrak{h}^*$ put

$$\mathfrak{g}_\lambda = \{x \in \mathfrak{g} : [h, x] = \lambda(h)x \text{ for all } h \in \mathfrak{h}\}.$$

A *root* is a nonzero λ with $\mathfrak{g}_\lambda \neq 0$. Write R for the finite set of roots.

From Lecture 7 we use the root decomposition

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha.$$

We also use the bracket rule and orthogonality. For all $\lambda, \mu \in \mathfrak{h}^*$,

$$[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subseteq \mathfrak{g}_{\lambda+\mu}, \qquad \kappa(\mathfrak{g}_\lambda, \mathfrak{g}_\mu) = 0 \quad \text{unless } \lambda + \mu = 0.$$

The Cartan subalgebra is abelian. The single cited import is the following.

Theorem 8.1 (Self-centraliser theorem · *C*)**.** *Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$, and let $\mathfrak{h}$ be a maximal toral subalgebra. Then*

$$\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

Source: Humphreys, Section 8.2, Proposition.

The other tools are Lie's theorem (Lecture 3), nondegeneracy of the Killing form (Lecture 4), complete reducibility (Lecture 5) and the $\mathfrak{sl}_2$-modules V_m (Lecture 6).

Lemma 8.2 (First consequences · *P*)**.** *(a) The restriction of κ to $\mathfrak{h}$ is nondegenerate.*
(b) For $\alpha \in R$, the pairing $\kappa\colon \mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}$ is nondegenerate. In particular, $-\alpha \in R$.

(c) $Z(\mathfrak{g}) = 0$, *and* R *spans* $\mathfrak{h}^*$.

Proof. For $\mathfrak{g} = 0$ all three claims are immediate. Otherwise they are the orthogonality and root-spanning results proved in Lecture 7. That lecture also proved $Z(\mathfrak{g}) = 0$ from semisimplicity. ∎

By Lemma 8.2(a), each $\lambda \in \mathfrak{h}^*$ has a unique $t_\lambda \in \mathfrak{h}$ with

$$\kappa(t_\lambda, h) = \lambda(h) \quad \text{for all } h \in \mathfrak{h}.$$

The map $\lambda \mapsto t_\lambda$ is a linear isomorphism $\mathfrak{h}^* \to \mathfrak{h}$. Transport κ to $\mathfrak{h}^*$ by

$$(\lambda, \mu) = \kappa(t_\lambda, t_\mu) = \lambda(t_\mu) = \mu(t_\lambda).$$

This is a nondegenerate symmetric complex bilinear form.

8.2 The triple attached to a root

Lemma 8.3 (Opposite root vectors · *P*). *Let* $\alpha \in R$, $x \in \mathfrak{g}_\alpha$ *and* $y \in \mathfrak{g}_{-\alpha}$. *Then*

$$[x, y] = \kappa(x, y)\, t_\alpha.$$

Proof. The bracket rule and the self-centraliser theorem give $[x, y] \in \mathfrak{g}_0 = \mathfrak{h}$. For $h \in \mathfrak{h}$, invariance of κ gives

$$\kappa(h, [x, y]) = \kappa([h, x], y) = \alpha(h)\,\kappa(x, y) = \kappa\big(h, \kappa(x, y)\, t_\alpha\big).$$

Both $[x, y]$ and $\kappa(x, y)t_\alpha$ lie in $\mathfrak{h}$. By nondegeneracy on $\mathfrak{h}$ (Lemma 8.2(a)), they are equal. ∎

Lemma 8.4 (Roots are not isotropic · *P*). *For every* $\alpha \in R$,

$$(\alpha, \alpha) = \alpha(t_\alpha) \neq 0.$$

Proof. Suppose $\alpha(t_\alpha) = 0$. By Lemma 8.2(b), choose $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ with $\kappa(x, y) = 1$. Put $s = [x, y]$. Lemma 8.3 gives $s = t_\alpha$, so

$$[s, x] = \alpha(s)x = 0, \qquad [s, y] = -\alpha(s)y = 0.$$

Thus $S = \operatorname{span}\{x, y, s\}$ is a subalgebra with $[S, S] = \mathbb{C}s$ and $[[S, S], [S, S]] = 0$. It is solvable.

The image $\operatorname{ad} S \subseteq \mathfrak{gl}(\mathfrak{g})$ is solvable as well. By Lie's theorem there is a basis of $\mathfrak{g}$ in which every $\operatorname{ad} z$, $z \in S$, is upper triangular. Then

$$\operatorname{ad} s = [\operatorname{ad} x, \operatorname{ad} y]$$

is strictly upper triangular, hence nilpotent.

On the other hand $s \in \mathfrak{h}$. By the root decomposition, $\operatorname{ad} s$ acts on $\mathfrak{g}_\gamma$ by $\gamma(s)$ and on $\mathfrak{h}$ by 0. So $\operatorname{ad} s$ is diagonalisable. A diagonalisable nilpotent operator is zero, so $s \in Z(\mathfrak{g}) = 0$. This contradicts $t_\alpha \neq 0$, which holds since $\alpha \neq 0$. ∎

Definition 8.5 (Coroot and root triple). Let $\alpha \in R$. The *coroot* of α is

$$h_\alpha = \alpha^\vee = \frac{2\, t_\alpha}{(\alpha, \alpha)} \in \mathfrak{h}.$$

For $\lambda \in \mathfrak{h}^*$ write $\langle \lambda, \alpha^\vee \rangle = \lambda(h_\alpha)$. A *root triple* for α is a triple $(x_\alpha, h_\alpha, y_\alpha)$ with $x_\alpha \in \mathfrak{g}_\alpha$, $y_\alpha \in \mathfrak{g}_{-\alpha}$ and $[x_\alpha, y_\alpha] = h_\alpha$.

Root triples exist. Choose $x_\alpha \neq 0$ in $\mathfrak{g}_\alpha$. Lemma 8.2(b) gives $y \in \mathfrak{g}_{-\alpha}$ with $\kappa(x_\alpha, y) \neq 0$. Rescale y to y_α with

$$\kappa(x_\alpha, y_\alpha) = \frac{2}{(\alpha, \alpha)}.$$

Lemma 8.3 then gives $[x_\alpha, y_\alpha] = h_\alpha$.

Proposition 8.6 (The rank-one subalgebra · *P*)**.** *Let $(x_\alpha, h_\alpha, y_\alpha)$ be a root triple. Then*

$$\alpha(h_\alpha) = 2, \qquad [h_\alpha, x_\alpha] = 2x_\alpha, \qquad [h_\alpha, y_\alpha] = -2y_\alpha.$$

Hence $S_\alpha = \operatorname{span}\{x_\alpha, h_\alpha, y_\alpha\}$ is a subalgebra, and $e \mapsto x_\alpha$, $h \mapsto h_\alpha$, $f \mapsto y_\alpha$ defines an isomorphism $\mathfrak{sl}_2 \to S_\alpha$. Moreover,

$$[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}h_\alpha,$$

and h_α is the only element of this line on which α takes the value 2.

Proof. First,

$$\alpha(h_\alpha) = \frac{2\,\alpha(t_\alpha)}{(\alpha, \alpha)} = 2.$$

Since $x_\alpha \in \mathfrak{g}_\alpha$, we get $[h_\alpha, x_\alpha] = \alpha(h_\alpha)x_\alpha = 2x_\alpha$. Similarly $[h_\alpha, y_\alpha] = -2y_\alpha$.

The relation $[x_\alpha, y_\alpha] = h_\alpha \neq 0$ forces $x_\alpha \neq 0$ and $y_\alpha \neq 0$. The three vectors lie in the independent summands $\mathfrak{g}_\alpha$, $\mathfrak{h}$ and $\mathfrak{g}_{-\alpha}$, so they are linearly independent. The linear map $\mathfrak{sl}_2 \to S_\alpha$ is therefore bijective. It respects the three defining brackets of $\mathfrak{sl}_2$, which determine all brackets. So it is an isomorphism onto S_α.

Finally, Lemma 8.3 gives $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathbb{C}t_\alpha = \mathbb{C}h_\alpha$, and the line contains h_α. For $c \in \mathbb{C}$, $\alpha(ch_\alpha) = 2c$, which equals 2 only for $c = 1$. ∎

Lecture 6 for S_α

Through the isomorphism, every finite-dimensional $\mathfrak{g}$-module V, such as $\mathfrak{g}$ under ad, is an S_α-module. Write

$$V(k) = \{v \in V : h_\alpha v = kv\}.$$

By Lecture 5 and Lecture 6, V is a direct sum of modules V_m. In V_m the basis vectors $v_j = f^j v_0$, $0 \leq j \leq m$, satisfy

$$hv_j = (m - 2j)v_j, \qquad ev_j = j(m - j + 1)v_{j-1}.$$

We use three consequences.

(W1) h_α is diagonalisable on V, with integer eigenvalues.

(W2) The number of summands V_m equals $\dim V(0) + \dim V(1)$. Indeed, each V_m contains exactly one of the eigenvalues $0, 1$, with multiplicity one.

(W3) In V_m the eigenvalues are $m, m-2, \ldots, -m$, each of multiplicity one. For $-m \leq k < m$, the operator x_α maps $V_m(k)$ isomorphically onto $V_m(k+2)$, since $j(m - j + 1) \neq 0$ for $1 \leq j \leq m$.

8.3 Root spaces are lines

Theorem 8.7 (Dimension one and reducedness · *P*)**.** *Let $\alpha \in R$.*

(a) $\dim \mathfrak{g}_\alpha = 1$.

(b) If $c \in \mathbb{C}$ and $c\alpha \in R$, then $c = \pm 1$.

Proof. Fix a root triple for α. On $\mathfrak{g}_{c\alpha}$ the coroot acts by $c\alpha(h_\alpha) = 2c$. Consider

$$M = \mathfrak{h} \oplus \bigoplus_{c\neq 0,\ c\alpha\in R} \mathfrak{g}_{c\alpha}.$$

Step 1: M is an S_α-submodule. The operator $\operatorname{ad} h_\alpha$ preserves each summand. By the bracket rule, $\operatorname{ad} x_\alpha$ sends $\mathfrak{g}_{c\alpha}$ into $\mathfrak{g}_{(c+1)\alpha}$. This is $\mathfrak{h}$ when $c = -1$, and otherwise it is zero or a summand of M. Also $\operatorname{ad} x_\alpha$ sends $\mathfrak{h}$ into $\mathfrak{g}_\alpha$. The argument for $\operatorname{ad} y_\alpha$ is the same.

Step 2: two known submodules. The subalgebra $S_\alpha \subseteq M$ is a submodule isomorphic to V_2. The hyperplane $\ker\alpha \subseteq \mathfrak{h}$ is a trivial submodule: for $k \in \ker\alpha$,

$$[x_\alpha, k] = -\alpha(k)x_\alpha = 0, \qquad [y_\alpha, k] = \alpha(k)y_\alpha = 0, \qquad [h_\alpha, k] = 0.$$

Since $\alpha(h_\alpha) = 2$, we have $\mathfrak{h} = \ker\alpha \oplus \mathbb{C}h_\alpha$ and $\ker\alpha \cap S_\alpha = 0$. Complete reducibility gives a submodule M' with

$$M = \ker\alpha \oplus S_\alpha \oplus M'.$$

Step 3: M' has only odd eigenvalues. The eigenvalue 0 occurs on M only for $c = 0$. Hence

$$M(0) = \mathfrak{h} = \ker\alpha \oplus \mathbb{C}h_\alpha,$$

and $M'(0) = 0$. Every V_m with m even has the eigenvalue 0. So every summand of M' is some V_m with m odd.

Step 4: $2\alpha \notin R$. The eigenspace $M(4)$ is $\mathfrak{g}_{2\alpha}$. The summands $\ker\alpha$ and S_α have eigenvalues $0, \pm 2$ only, and M' has odd eigenvalues. Thus $M(4) = 0$, and $2\alpha \notin R$. This holds for every root.

Step 5: $M' = 0$. If $\alpha/2$ were a root, Step 4 applied to $\alpha/2$ would give $\alpha \notin R$. So $\alpha/2 \notin R$. The eigenvalue 1 on M could only come from $c = 1/2$, hence

$$M(1) = 0.$$

Every V_m with m odd has the eigenvalue 1. Hence $M' = 0$.

Conclusion. Now $M = \ker\alpha \oplus S_\alpha$, with eigenvalues $0, \pm 2$ only. For $c \neq 0$ this forces $2c = \pm 2$, which is (b). The eigenspace $M(2)$ equals $\mathfrak{g}_\alpha$ and also $\mathbb{C}x_\alpha$. This is (a). ∎

8.4 Root strings and integrality

Definition 8.8 (Root string)**.** Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$. The *α-string through β* is the set of roots of the form $\beta + i\alpha$ with $i \in \mathbb{Z}$.

Theorem 8.9 (Root strings · *P*). *Let $\alpha, \beta \in R$ with $\beta \neq \pm\alpha$.*

(a) $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$.

(b) There are integers $r, q \geq 0$ such that, for $i \in \mathbb{Z}$,

$$\beta + i\alpha \in R \iff -r \leq i \leq q.$$

(c) These integers satisfy

$$r - q = \langle \beta, \alpha^\vee \rangle.$$

(d) If $\alpha + \beta \in R$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$.

Since $\langle \pm\alpha, \alpha^\vee \rangle = \pm 2$, part (a) makes every $\langle \beta, \alpha^\vee \rangle$ with $\alpha, \beta \in R$ an integer, a *Cartan integer*.

Proof. Fix a root triple for α and put

$$K = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_{\beta+i\alpha}.$$

No $\beta + i\alpha$ is zero. Otherwise $\beta = -i\alpha$ would be a root proportional to α, and Theorem 8.7(b) would give $\beta = \pm\alpha$. By the bracket rule, $\operatorname{ad} x_\alpha$ and $\operatorname{ad} y_\alpha$ move each summand to a neighbouring summand, never into $\mathfrak{h}$. Thus K is an S_α-submodule. It is nonzero because $\mathfrak{g}_\beta \neq 0$.

The coroot acts on $\mathfrak{g}_{\beta+i\alpha}$ by

$$(\beta + i\alpha)(h_\alpha) = \langle \beta, \alpha^\vee \rangle + 2i.$$

By (W1) this is an integer. The case $i = 0$ gives (a).

By Theorem 8.7(a), each summand has dimension at most one. The eigenvalues $\langle \beta, \alpha^\vee \rangle + 2i$ are distinct and have the same parity. Hence $\dim K(0) + \dim K(1) \leq 1$, and (W2) shows that K is irreducible, say $K \cong V_m$. By (W3), the eigenvalues on K are exactly $m, m-2, \ldots, -m$. The value for $i = 0$ is one of them, so every value $\langle \beta, \alpha^\vee \rangle + 2i$ has the parity of m. Hence $\beta + i\alpha \in R$ exactly when

$$-m \leq \langle \beta, \alpha^\vee \rangle + 2i \leq m.$$

These i form an interval of integers containing 0. Call it $[-r, q]$. This is (b).

The endpoints give the highest and lowest eigenvalues:

$$\langle \beta, \alpha^\vee \rangle + 2q = m, \qquad \langle \beta, \alpha^\vee \rangle - 2r = -m.$$

Adding the two equations gives $2\langle \beta, \alpha^\vee \rangle = 2r - 2q$, which is (c).

For (d), $\alpha + \beta \in R$ means $q \geq 1$. Then $k = \langle \beta, \alpha^\vee \rangle = m - 2q < m$, and $k \geq -m$. By (W3), $\operatorname{ad} x_\alpha$ maps $\mathfrak{g}_\beta = K(k)$ isomorphically onto $K(k+2) = \mathfrak{g}_{\alpha+\beta}$. Since $\mathfrak{g}_\alpha = \mathbb{C}x_\alpha$ by Theorem 8.7(a),

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = [x_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}.$$

■

For $\alpha \in R$, define the linear map

$$s_\alpha : \mathfrak{h}^* \longrightarrow \mathfrak{h}^*, \qquad s_\alpha(\lambda) = \lambda - \lambda(h_\alpha)\alpha.$$

Since $\alpha(h_\alpha) = 2$, we have $s_\alpha(\lambda)(h_\alpha) = -\lambda(h_\alpha)$. Applying the formula again gives $s_\alpha^2 = \mathrm{id}$.

Corollary 8.10 (Reflections permute the roots · *P*). *For $\alpha, \beta \in R$,*

$$s_\alpha(\beta) = \beta - \langle \beta, \alpha^\vee \rangle \alpha \in R.$$

Proof. For $\beta = \pm\alpha$ the element is $\mp\alpha$, a root by Lemma 8.2(b). Otherwise it equals $\beta + (q-r)\alpha$, which is a root because $-r \leq q - r \leq q$. Thus $s_\alpha(R) \subseteq R$. Since $s_\alpha^2 = \mathrm{id}$, equality holds. ■

8.5 The real span of the roots

Lemma 8.11 (The Killing form on $\mathfrak{h} \cdot P$)**.** *For $h, h' \in \mathfrak{h}$ and $\lambda, \mu \in \mathfrak{h}^*$,*

$$\kappa(h, h') = \sum_{\gamma \in R} \gamma(h)\gamma(h'), \qquad (\lambda, \mu) = \sum_{\gamma \in R} (\gamma, \lambda)(\gamma, \mu).$$

Proof. The operator $\operatorname{ad} h \operatorname{ad} h'$ is 0 on $\mathfrak{h}$ and multiplication by $\gamma(h)\gamma(h')$ on the line $\mathfrak{g}_\gamma$. Take its trace. For the second formula, put $h = t_\lambda$, $h' = t_\mu$ and use $\gamma(t_\lambda) = (\gamma, \lambda)$. ∎

Theorem 8.12 (The Euclidean space of roots $\cdot$ *P*)**.** *(a) For $\alpha, \beta \in R$, $(\alpha, \beta) \in \mathbb{Q}$ and $(\alpha, \alpha) > 0$.*

(b) Let E be the real span of R in $\mathfrak{h}^*$. Then $\dim_{\mathbb{R}} E = \dim_{\mathbb{C}} \mathfrak{h}^*$ and $\mathfrak{h}^* = E \oplus iE$.

(c) The form $(\ ,\ )$ is real-valued and positive definite on E.

Proof. (a) Lemma 8.11 gives $(\beta, \beta) = \sum_\gamma (\gamma, \beta)^2$. Divide by $(\beta, \beta)^2$, which is nonzero by Lemma 8.4. Since $\langle \gamma, \beta^\vee \rangle = 2(\gamma, \beta)/(\beta, \beta)$,

$$\frac{1}{(\beta, \beta)} = \sum_{\gamma \in R} \Big(\frac{(\gamma, \beta)}{(\beta, \beta)} \Big)^2 = \frac{1}{4} \sum_{\gamma \in R} \langle \gamma, \beta^\vee \rangle^2.$$

The right side is a quarter of a sum of squares of integers, and its term $\gamma = \beta$ equals 1. So (β, β) is a positive rational. Then $(\gamma, \beta) = \frac{1}{2}\langle \gamma, \beta^\vee \rangle (\beta, \beta)$ is rational.

(b) By Lemma 8.2(c), choose roots $\alpha_1, \dots, \alpha_\ell$ forming a basis of $\mathfrak{h}^*$. Their Gram matrix $G = ((\alpha_i, \alpha_j))$ is rational by (a) and invertible, because the form is nondegenerate. For $\beta \in R$ write $\beta = \sum_i c_i \alpha_i$ with $c_i \in \mathbb{C}$. Then

$$(\beta, \alpha_j) = \sum_{i=1}^{\ell} c_i (\alpha_i, \alpha_j) \quad (1 \le j \le \ell).$$

The left sides are rational, so inverting G shows that every c_i is rational. Thus $E = \mathbb{R}\alpha_1 \oplus \cdots \oplus \mathbb{R}\alpha_\ell$. Since the α_i form a $\mathbb{C}$-basis, $\mathfrak{h}^* = E \oplus iE$.

(c) The form is bilinear and, by (a), real on pairs of roots. Since E is the real span of R, the value (λ, μ) is real for all $\lambda, \mu \in E$. For $\lambda \in E$, Lemma 8.11 gives

$$(\lambda, \lambda) = \sum_{\gamma \in R} (\gamma, \lambda)^2 \ge 0.$$

If equality holds, then $(\gamma, \lambda) = 0$ for every root. Since R spans $\mathfrak{h}^*$ and the form is nondegenerate, $\lambda = 0$. ∎

For $\alpha \in R$ and $\lambda \in E$, the reflection formula becomes

$$s_\alpha(\lambda) = \lambda - \frac{2(\lambda, \alpha)}{(\alpha, \alpha)}\, \alpha.$$

The coefficient is real, so s_α maps E to itself.

Positive definiteness gives an orthogonal decomposition $E = \mathbb{R}\alpha \oplus \alpha^\perp$. The map s_α fixes $\alpha^\perp$ and negates α, so it is an orthogonal reflection of E. By Corollary 8.10, it permutes R.

The *Weyl group* $W \subseteq \mathrm{GL}(E)$ is generated by all s_α. The form on E is W-invariant. Lecture 9 studies W.

Remark 8.13 (The complex Killing form is not Euclidean)**.** The form is positive definite on the real span E. On $\mathfrak{h}^*$ it is complex bilinear, so $(i\lambda, i\lambda) = -(\lambda, \lambda)$. On $\mathfrak{g}$ itself, since $2\alpha \neq 0$, orthogonality gives $\kappa(x_\alpha, x_\alpha) = 0$ although $x_\alpha \neq 0$.

8.6 Root strings in A_2 and C_2

Write E_{ij} for matrix units, and ε_i for the i-th diagonal entry as a linear form on diagonal matrices. For roots α_1, α_2 with coroots h_1, h_2 put

$$a_{ij} = \alpha_j(h_i),$$

as in the course convention. Lecture 9 shows that the pairs below are simple roots.

Example 8.14 ($A_2 = \mathfrak{sl}_3$)**.** Use the traceless diagonal Cartan subalgebra of $\mathfrak{sl}_3$ from Lecture 7. Its roots are $\varepsilon_i - \varepsilon_j$ for $i \neq j$. For $\alpha = \varepsilon_i - \varepsilon_j$,

$$[E_{ij}, E_{ji}] = E_{ii} - E_{jj}, \qquad \alpha(E_{ii} - E_{jj}) = 2.$$

By Proposition 8.6, $h_\alpha = E_{ii} - E_{jj}$.

Put $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$. Test the α_1-string through α_2:

$$\begin{aligned} \alpha_2 - \alpha_1 &= -\varepsilon_1 + 2\varepsilon_2 - \varepsilon_3 \notin R, \\ \alpha_2 + \alpha_1 &= \varepsilon_1 - \varepsilon_3 \in R, \\ \alpha_2 + 2\alpha_1 &= 2\varepsilon_1 - \varepsilon_2 - \varepsilon_3 \notin R. \end{aligned}$$

The coefficient vectors in this test all have sum zero. By Exercise 7.1, equality on $\mathfrak{h}$ would force equality of these vectors. The two excluded forms are therefore not roots. Thus $r = 0$ and $q = 1$. The string theorem gives

$$a_{12} = \alpha_2(h_1) = r - q = -1.$$

Directly, $\alpha_2(E_{11} - E_{22}) = -1 - 0 = -1$. Exchanging α_1 and α_2 gives $a_{21} = -1$.

Example 8.15 ($C_2 = \mathfrak{sp}_4$)**.** Let

$$J = \begin{pmatrix} 0 & I_2 \\ -I_2 & 0 \end{pmatrix}, \qquad \mathfrak{sp}_4 = \{X \in \mathfrak{gl}_4(\mathbb{C}) : X^{\mathsf{T}} J + JX = 0\}.$$

Solving the condition in 2×2 blocks gives

$$X = \begin{pmatrix} A & B \\ C & -A^{\mathsf{T}} \end{pmatrix}, \qquad B = B^{\mathsf{T}}, \quad C = C^{\mathsf{T}}.$$

Exercise 8.1 proves that $\mathfrak{sp}_4$ is semisimple and that the matrices $\operatorname{diag}(a_1, a_2, -a_1, -a_2)$ form a Cartan subalgebra $\mathfrak{h}$. Computing $[h, X]$ gives the roots and root vectors below. In each line, the first vector belongs to the plus sign:

$$\begin{aligned} \pm(\varepsilon_1 - \varepsilon_2): &\quad E_{12} - E_{43}, \;\; E_{21} - E_{34}, \\ \pm(\varepsilon_1 + \varepsilon_2): &\quad E_{14} + E_{23}, \;\; E_{41} + E_{32}, \\ \pm 2\varepsilon_1: &\quad E_{13}, \;\; E_{31}, \\ \pm 2\varepsilon_2: &\quad E_{24}, \;\; E_{42}. \end{aligned}$$

Put $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = 2\varepsilon_2$. Brackets of opposite root vectors give

$$\begin{aligned} [E_{12} - E_{43}, E_{21} - E_{34}] &= E_{11} - E_{22} - E_{33} + E_{44}, \\ [E_{24}, E_{42}] &= E_{22} - E_{44}. \end{aligned}$$

Each root takes the value 2 on the corresponding right side. By Proposition 8.6,

$$h_1 = \operatorname{diag}(1,-1,-1,1), \qquad h_2 = \operatorname{diag}(0,1,0,-1).$$

The α_1-string through α_2 is found by testing:

$$\begin{aligned} \alpha_2 - \alpha_1 &= -\varepsilon_1 + 3\varepsilon_2 \notin R,\\ \alpha_2 + \alpha_1 &= \varepsilon_1 + \varepsilon_2 \in R,\\ \alpha_2 + 2\alpha_1 &= 2\varepsilon_1 \in R,\\ \alpha_2 + 3\alpha_1 &= 3\varepsilon_1 - \varepsilon_2 \notin R. \end{aligned}$$

Thus $r = 0$ and $q = 2$, and

$$a_{12} = \alpha_2(h_1) = r - q = -2.$$

This agrees with the direct value $2\varepsilon_2(h_1) = 2 \cdot (-1)$.

The α_2-string through α_1 is shorter:

$$\begin{aligned} \alpha_1 - \alpha_2 &= \varepsilon_1 - 3\varepsilon_2 \notin R,\\ \alpha_1 + \alpha_2 &= \varepsilon_1 + \varepsilon_2 \in R,\\ \alpha_1 + 2\alpha_2 &= \varepsilon_1 + 3\varepsilon_2 \notin R. \end{aligned}$$

So $r = 0$, $q = 1$ and $a_{21} = \alpha_1(h_2) = -1$. The Cartan matrix of this pair is

$$(a_{ij}) = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}.$$

The asymmetry of this matrix records root lengths. By definition, $a_{ij} = \alpha_j(h_i) = 2(\alpha_j,\alpha_i)/(\alpha_i,\alpha_i)$. Since $(\alpha_1,\alpha_2) \neq 0$,

$$\frac{a_{12}}{a_{21}} = \frac{(\alpha_2,\alpha_2)}{(\alpha_1,\alpha_1)} = 2.$$

So α_2 is long and α_1 is short. Reflections transport lengths:

$$s_{\alpha_2}(\alpha_1) = \alpha_1 - a_{21}\alpha_2 = \varepsilon_1 + \varepsilon_2, \qquad s_{\alpha_1}(\alpha_2) = \alpha_2 - a_{12}\alpha_1 = 2\varepsilon_1.$$

By W-invariance, the roots $\pm\varepsilon_1 \pm \varepsilon_2$ are short and the roots $\pm 2\varepsilon_i$ are long.

Self-checks

(1) In $\mathfrak{sl}_3$, find the $(\varepsilon_1 - \varepsilon_2)$-string through $\varepsilon_1 - \varepsilon_3$, and the value of $\varepsilon_1 - \varepsilon_3$ on $h_{\varepsilon_1-\varepsilon_2}$.

(2) Why is the Cartan matrix of C_2 not symmetric, while that of A_2 is?

(3) For $\alpha \in R$, compute $\kappa(ih_\alpha, ih_\alpha)$.

Answers. (1) Put $\alpha = \varepsilon_1 - \varepsilon_2$ and $\beta = \varepsilon_1 - \varepsilon_3$. Then $\beta - \alpha = \varepsilon_2 - \varepsilon_3 \in R$, while $\beta - 2\alpha$ and $\beta + \alpha$ are not roots, by the same coefficient test. So $r = 1$, $q = 0$ and the value is 1. (2) The ratio a_{12}/a_{21} equals a ratio of squared lengths. It is 2 in C_2 and 1 in A_2. (3) Since $\kappa(t_\alpha, t_\alpha) = (\alpha,\alpha) > 0$,

$$\kappa(ih_\alpha, ih_\alpha) = -\frac{4\,\kappa(t_\alpha,t_\alpha)}{(\alpha,\alpha)^2} = -\frac{4}{(\alpha,\alpha)} < 0.$$

Summary

Each root α gives $S_\alpha \cong \mathfrak{sl}_2$. Viewing $\mathfrak{g}$ as an S_α-module shows that root spaces are lines and that $\pm\alpha$ are the only roots proportional to α.

For $\beta \neq \pm\alpha$, the root spaces of the α-string through β span an irreducible S_α-module, and $r - q = \langle \beta, \alpha^\vee \rangle$. The real span E of the roots carries a positive definite W-invariant form. Lecture 9 axiomatises this structure.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 8.3–8.5. Erdmann–Wildon, *Introduction to Lie Algebras*, Sections 10.4–10.6.